

Improving Effectiveness of Sales Initiatives in a Retail Bank

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Understand the Business Requirement

Retail bank often spend considerable amount of resources and time in acquiring customers and new business from existing customers. This case pertains to one such mid-sized bank which wanted to increase its market share in the personal loan space. It was observed that the return on various initiatives to cross sell were limited and were becoming burden on the operational cost.

As per the BMGI methodology we start with deep dive into the business requirement through structured questions

- What's Happening – Which segment of customer are responding positively to the sales campaigns and availing loan.
- Why are customers in these segments responding positively and others are not - What drives the customer decision.
- What will happen If our marketing campaigns and cross selling efforts are focused only towards the customers who have higher probability of taking loan
- How can we make our sales effort more effective or in simple terms improve the hit rate (Number of customers took loan/Number of customers contacted)?

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We also systematically engage the key stakeholders, articulate the success criteria, define scope of the project/program, define timeline, address the data privacy requirements and set up a governance.

Obtain Data

Data available in multiple systems within and out-side bank were combined to OBTAIN the information related to

Credit Card Spend, Annual Income, Home Loan Aailed and Amount etc.

Personal Information (Age, Experience, Number of Family Members, ZIP Code, Education Qualification)

It was decided not to include Gender and Relationship Status in the study.

Overall a sample of 5000 customers was extracted with the information as given in Table 1,

Table 1 : Meta Model of the Data Extracted

ID	Age	Experience	Income	ZIP Code	Family	CCAvg	Education	Mortgage	Personal Loan	Securities Account	CD Ac.	Online	Credit Card
1	25	1	49	91107	4	1.60	1	0	0	1	0	0	0
2	45	19	34	90089	3	1.50	1	0	0	1	0	0	0
3	39	15	11	94720	1	1.00	1	0	0	0	0	0	0
4	35	9	100	94112	1	2.70	2	0	0	0	0	0	0
5	35	8	45	91330	4	1.00	2	0	0	0	0	0	1
6	37	13	29	92121	4	0.40	2	155	0	0	0	1	0
7	53	27	72	91711	2	1.50	2	0	0	0	0	1	0
8	50	24	22	93943	1	0.30	3	0	0	0	0	0	1
9	35	10	81	90089	3	0.60	2	104	0	0	0	1	0
10	34	9	180	93023	1	8.90	3	0	1	0	0	0	0

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Treatment of Duplicates and Missing Values

ZIP Codes	Education			Grand Total
	1	2	3	
9307			1	1
90007	5		1	6
90009	4	2	2	8
90011	1	1	1	3
90016			2	2
90018	2		2	4
90019	1	2	2	5
90024	20	12	18	50
90025	5	7	7	19
90027	2		1	3
90028	4	2	2	8
90029	4	1		5
90032	5	2	1	8
90033	4	2	3	9
90034	12	2	6	20
90035	1	1	4	6
90036	5	1	1	7
90037	2	1	2	5
90041	5	1	4	10
90044	1		1	2
90045	3			3

As the data was extracted from the active business data basis there were no duplicate records

- There were missing values, for example certain ZIP Codes had customers with post-graduation and few didn't have any post-graduate
- Each instance of missing value for investigated and discussed with the domain expert before treatment them appropriately to ensure the accuracy of inferences drawn

The missing values can occur due to various reasons. Depending on the category a given missing value belong to there are ways treat the same.

Missing Not at Random (MNAR) – As it is seen here that certain ZIP could do not have individuals with certain qualifications. The same can happen with income levels too. It is advised to treat these values in consultation with domain experts. The approach adapted in this case take care of these type of missing values.

Missing at Random (MAR) is very difficult to established must be substantiated with business logic. While in this case we didn't have any such situation, a simple example could be males are less likely to fill the survey questionnaire completely, but this might not have bearing on their satisfaction level.

Missing completely at Random (MCAR) – The proportion of missing values does depend on specific category of response and there's no relationship between with any of the features in the data set, these can be Logically imputed.

Identification of Data Type: While it might sound trivial, identification of data in right category is of utmost importance, a Variable and Attribute type can be easily distinguished, but there are frequent instances of misidentification of Ordinal & Categorical data. Assigning a categorical variable (for example ZIP Code) as Ordinal can result in misleading conclusions, similarly an Ordinal data (Number of family members) considered as categorical also have risk of high error in conclusions.

Explore

Simple One Way and Two-Way Analysis - One-way analysis by Number or family members and Education indicate that the hit rates are pretty low 7-8% for customers with 1 or to family members and 4% for Non-Graduates. However, when we do a simple two-way analysis it shows for Non-Graduates with family size 3 and 4, the hit rate is observed to be 10-11% significant enough to focus on the category. Considering the current day complex and dynamic business scenarios we don't deal with only 2,3,4 variables rather the number of variables could be 10, 20, 30 or even more. This is one of the many other reasons that we need to deploy some of the analytics tools to get reliable insights. In this particular case we worked only with 10-12 variables as the domain experts were able to give fairly high amount on understanding on the variables to be considered.

At this stage, we also looked at the extreme values or outliers and appropriately treat them to avoid any bias.

Table 2 : Hit Rate by Number of Family Members

No of Family Member	% Loan Aailed	
	No	Yes
1	93%	7%
2	92%	8%
3	87%	13%
4	89%	11%
Grand Total	90%	10%

Table 3 : Hit Rate by Qualification

Education	% Loan Aailed	
	No	Yes
Non Graduate	96%	4%
Graduate	87%	13%
Post Graduate	86%	14%
Grand Total	90%	10%

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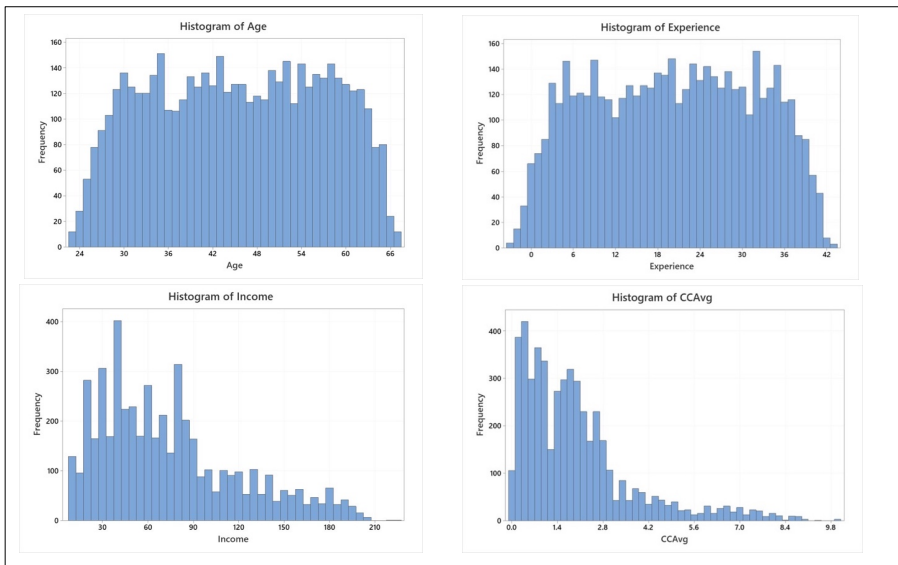
Table 4 : Hit rate by Qualification and Number of family Members

% Avail Loan	Family Size				
Education	1	2	3	4	Overall
Non Graduate	1%	1%	11%	10%	4%
Graduate	12%	19%	11%	11%	13%
Post Graduate	12%	14%	18%	12%	14%
Overall	7%	8%	13%	11%	10%

Data Distributions - Other dimensions we need to explore are the shape, the scale and identify the extreme values or what are known as outlier. In past statistical & quality methodologies have depended heavily on normal distributions which hardly occurs in the current business environment.

As a case in point, Age and Experience follows almost flat or Uniform distribution where-as Income and Credit card follow right skewed distribution. With this it makes it necessary for us to choose the techniques which are agnostic to the underlying distribution of the data sets.

Graph 1 : Distributions for Age, Experience, Salary and Average spend on credit card

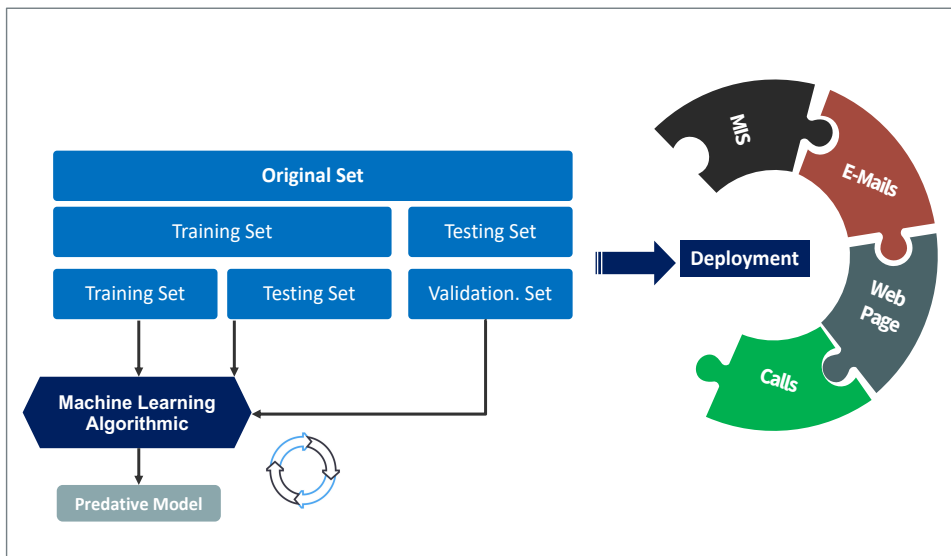


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The key differentiator in applying Analytics Framework comes from the fact that we create an algorithm which learns to generates innumerable models using a portion of the data (60-70), then test it on a smaller portion (20-30%) and finally validate. While the majority of work is done by the machine, the project leader and team need to ensure that the selected model is the right fit for the defined business objective.

Figure 1 : Predictive model

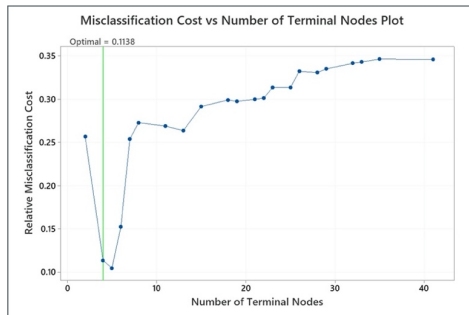


Classification & Regression - Tree Cost of Misclassification (Error Rate)

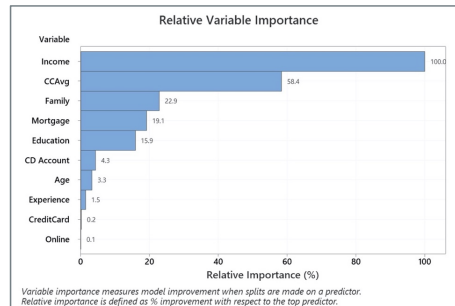
As the hit rate in percentage (%) is continuous data, we deployed Regression tree. The algorithm created around 40 models with varying number of splits known as nodes. With each of the model cost of misclassification (error rate) was computed and model within minimal error rate that is around 11% was selected.

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Graph 2 : Cost of Misclassification (Error Rate)



Graph 3 : Relative Importance (Place side by side)



Interpret

Decision Tree

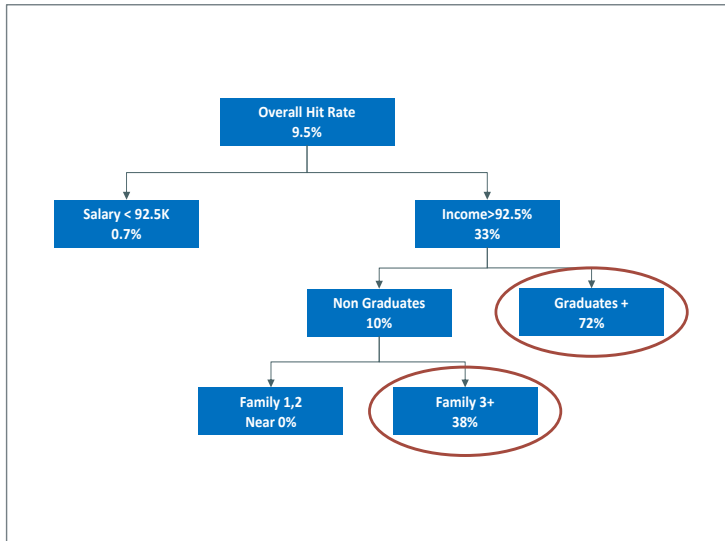
As it can be seen in Figure 2, decision tree provides the output in a manner as any human brain will think. The tool has its strength in providing

- (a) Relative importance of the factors (Graph 3)
- (b) Multiple alternatives in terms of combinations of important factors
- (c) Prediction of performance under the set of combinations

Figure 2 shows that **customers with Income >92.5K and Graduates + - Hit Rate 72%** and **Non-Graduates - with family size of 3+ the category with hit rate of 38%**

This reduces our effort in terms of directing our time and resources on part of the customer segment which is likely to result in better hit-rate.

Figure 2: Decision Tree for Hit Rate by Salary, Education and Number of Family Members



Prediction

Using the model, we predict the values as given in Table on page 5. It can be seen that for the customers with same salary range and qualification the hit rate improves from <1%. To 72% as the family members increase to 3 or more.

Table 5 : Prediction @ Select segments of Customers

Income	Education	Family	Hit Rate
75	Post-Graduate	4	0.7%
150	Non-Graduate	1	0.8%
150	Non-Graduate	2	0.8%
150	Non-Graduate	3	72%
150	Non-Graduate	4	72%

Table 6 – Confusion Matrix

Confusion Matrix

Actual Class Count	Predicted Class (Training)			Predicted Class (Test)			
	1	0	%Correct	1	0	%Correct	
1 (Event)	480	449	31	93.5	446	34	92.9
0	4520	200	4320	95.6	194	4326	95.7
All	5000	649	4351	95.4	640	4360	95.4
Statistics				Training (%)		Test (%)	
True positive rate (sensitivity or power)				93.5		92.9	
False positive rate (type I error)				4.4		4.3	
False negative rate (type II error)				6.5		7.1	
True negative rate (specificity)				95.6		95.7	

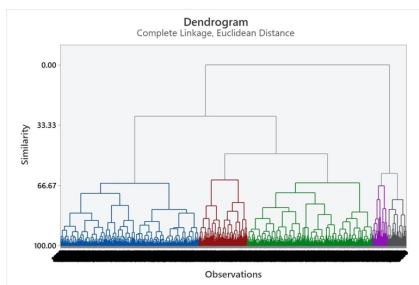
Confusion Matrix – Finally we access the chances of wrongly categorizing the customer. Table 6 shows that there is 93.5% (True Positive) accuracy in correctly identifying a customer who is likely to take a loan. Also, the chances of identifying a customer who is unlikely to take loan are 95.6% (True Negative). These numbers are also comparable between the training and testing data sets. Hence these perditions were applied to launch the further campaigns.

Unsupervised Learning – Customer Segmentation

Businesses often come across situation where they want to segment the data to create New Products, Decide on Service Fee etc. and the impact of these changes in not available prior to making the decision.

One of the techniques used to accomplish this objective is cluster observations (or customers), which divides data into five logical clusters based on distances between their centers for varies underlying measurements. In this specific case the analysis resulted in 5 logical group of customers shown as Blue, Red, Green, Violet and Black in Figure 3. The detailed descriptions of these clusters are given in table 6.

Figure 3 : Customer Segmentation using key influencing factor



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Cluster-1 have 1998 customers and it constitutes account holders with average salary of 29.3K dollars per annum with credit card spend of 0.9K Dollars, whereas Cluster-4 and 5 have account holders with salaries of 188.6 & 149.6 with credit card spends of 4.7 & 4.0K USD. This is a rich information for the bank to leverage on various initiatives and decisions it takes to optimize the stakeholder's value.

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Table 7 : Cluster Analysis

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ID	Age	Experience	Income	ZIP Code	Family	CCAvg	Education	Mortgage	Securities Account	CD Account	Online	Credit Card
1	25	1	49	91107	4	1.60	1	0	1	0	0	0
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3	39	15	11	94720	1	1.00	1	0	0	0	0	0
4	35	9	100	94112	1	2.70	2	0	0	0	0	0
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6	37	13	29	92121	4	0.40	2	155	0	0	1	0
7	53	27	72	91711	2	1.50	2	0	0	0	1	0
8	50	24	22	93943	1	0.30	3	0	0	0	0	1
9	35	10	81	90089	3	0.60	2	104	0	0	1	0
10	34	9	180	93023	1	8.90	3	0	0	0	0	0

Feature	Cluster1	Cluster2	Cluster3	Cluster4	Cluster5	Total
No of Customers	1581	621	2037	177	584	5000
Age	43.5	46.1	46.7	42.2	45.7	45.3
Experience	18.2	21.0	21.4	17.0	20.7	20.1
Income	29.3	108.2	66.1	188.6	149.6	73.8
Family	2.5	2.1	2.6	1.9	2.0	2.4
CCAvg	0.9	2.6	1.7	4.7	4.0	1.9
Education	2.0	1.7	2.0	1.7	1.5	1.9

Interpretation from Cluster Analysis

It can be seen that 177 Customers in Cluster 4 and 584 Customers in Cluster 5 come from higher income group and have considerably higher spend to the extent of 4 times as compared to the 1581 Customers in Cluster 1. Specifically, in Cluster 4, customers are in lower age group with smaller family sizes. These are rich set of insights which are utilized for building new products and services, launching marketing initiatives, and many other areas.

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While this case focuses on Personal Loan, same approach can be used for Building and Testing New Products, Segmenting Employees, Segmenting Markets and various other business scenarios.

Point to Remember - Threat to Causal Inferences

Any causal inference has a threat of providing completely misleading information, one must keep cognizance the below biases

- (a) **Bias in Selection** is the bias introduced while selecting the sample groups or data in a manner which is not randomized.
- (b) **Simultaneity bias** occurs when there is interrelationship between the dependent and independent variable. Motivated employee create higher sales and get higher bonuses OR Higher Bonuses, Motivates employees and results in higher sales
- (c) **Omitted Variable Bias** occurs when the dependent variable is correlated with another variable which is not considered in the study. For example, smokers have higher health issues, the real reason could be that those who smoke, also consume alcohol and the real cause of health issue is the combination rather than one.
- (d) **Measurement Error** can occur due to scale, resolution or various other factors and depends largely on the system where the measurement is taken.

A statistically designed experiment is highly recommended where the risk of wrong conclusions is high, or the deployment involves higher amount of resources and time.